

State of charge estimation for lithium battery based on grey Kalman filter model

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Abstract: In this paper, a grey Kalman filter model is proposed for lithium battery charge state estimation. Firstly, this paper establishes a recursive relation equation between the front and back terms through the grey model (GM). Secondly, the state space expression is constructed based on the recursive relationship equation. Next, the Kalman filter algorithm is integrated to form a grey Kalman filter model. Finally, the charge state is estimated based on public lithium battery data. In this paper, the state of charge is estimated from three different aspects, including different driving cycles, randomly mixed driving cycles, and the estimation of the state of charge by different temperatures under the same driving cycle conditions. On this basis, the model is applied to a life scenario using the charge state of 20 electric vehicles. The results show that the proposed model has good accuracy.

Keywords: grey model (GM), lithium batteries, state of charge, electric vehicle.

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1. Introduction

Battery management systems (BMS) are essential for electric vehicles (EV), ensuring battery safety, enhancing performance, and prolonging lifespan. Accurate state of charge (SOC) estimation of lithium batteries directly affects key vehicle functions like range and power output. However, SOC estimation faces significant challenges due to the battery's complex internal mechanisms and its dependence on factors such as capacity, internal resistance, temperature, discharge rate, and aging. Chen et al. [1] improved the original capacity retention of capacitors by facile self-assembly of sandwich-like MXene/graphene oxide/NiMn layered double hydroxide nanocomposites. Additionally, variable driving condi-

tions like acceleration, deceleration, and slope changes further complicate SOC estimation. Consequently, this topic has attracted extensive research attention.

The SOC of a lithium battery represents the ratio of its remaining capacity to its fully charged capacity, expressed as a percentage [2]. A 100% SOC indicates a fully charged battery, while 0% means complete discharge [3]. As a critical parameter in BMS, SOC estimation helps users monitor remaining power and ensures safe, reliable operation of EV. For instance, He et al. [4] proposed an adaptive extended Kalman filter for SOC estimation based on an improved Thevenin model [4]. Current SOC estimation methods can be categorized into characterization parameter-based methods [5], ampere-time integration methods [6], model-based methods [7] and data-driven methods [8], etc. These approaches are summarized as follows.

(i) Method based on characterization parameters. This approach estimates SOC by establishing offline correlations between measurable parameters (e.g., remaining capacity, impedance spectrum, and open-circuit voltage) and battery state. Key studies include: based on the assumption of smoothness of the open circuit voltage (OCV)- t curve during the discharge process of lithium batteries. Xue et al. [9] realized the extraction of the OCV-SOC relationship for arbitrary current-voltage measurements by using a non-dominated sorting genetic algorithm, and then performed SOC verification in combination with extended Kalman filtering in the urban dynamometer driving schedule (UDDS) working condition. Xiao et al. [10] proposed an online OCV identification method, demonstrating accurate SOC estimation across wide temperature ranges with strong robustness to initial SOC errors. Zheng et al. [11] compared two OCV-SOC measurement techniques (low-current and incremental OCV tests) at multiple temperatures. The above scholars have established a characterization relationship between OCV and SOC. From these studies, it can be

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found that the characterization equation has a strong temperature dependence.

(ii) Ampere-hour integration method. The ampere-hour integration method is a technique for calculating battery capacity by integrating current over time [12]. Because the amount of power that can be discharged from a battery in practice is often lower than its nominal capacity due to internal resistance, the Coulomb efficiency η is added for correction [13]. The ampere-time integration method is a very classical method for estimating the state of charge. There are some drawbacks to the use of the ampere-time integral method, such as the difficulty of obtaining the exact value of its initial SOC. In general, the ampere-time integration method is often used in conjunction with other methods. For example, the initial SOC is determined using the open-circuit voltage OCV, and then the subsequent SOC trajectory is calculated using the anh-time integral method. Pan et al. [14] used the anh-time integral method and the equivalent circuit model to establish the state-space expression, and updated the state estimation using the Hermite interpolation interpolated open-circuit voltage model.

(iii) Model-based approach. The model-based approach utilizes a model and a state estimation algorithm to accomplish charge state estimation. Among them, Kalman filter-type algorithms are the most used methods in charge state estimation. Currently, Kalman filtering has been generalized from the original linear system to a nonlinear system, which is also known as extended Kalman filtering. Extended Kalman filtering uses Taylor expansion to linearize a nonlinear system. This method introduces truncation error and increases the charge state estimation error. Therefore, an improved Kalman filtering algorithm will be used to improve the accuracy of state estimation. Chen et al. [15] proposed an adaptive fractional order square root traceless Kalman filter to estimate the lithium-ion battery charge state. Besides, the power battery model can also be improved and optimized.

(iv) Data-driven methodology based approach. Data-driven approaches estimate battery SOC by analyzing voltage, current, and other operational data using machine learning models such as neural networks and support vector machines. Sesidhar et al. [16] classified these methods into six categories: classification and regression-based methods; fuzzy rule-based algorithms; combinatorial algorithms; probabilistic algorithms; recursive neural network; feedforward neural network algorithms. The advantage of these data-driven methods is that they can monitor SOC in real-time and do not require knowledge of the internal structure and chemical reaction mechanism of the battery. However, the data-driven method

also has some limitations. Firstly, this method requires a large amount of historical data for training and learning, and the model may need to be retrained for some new batteries or batteries under different usage scenarios. Secondly, the choice of data model and parameter settings also affect the accuracy and reliability of the estimation results, which need to be finely tuned and optimized.

In recent years, grey system theory models have been applied in the field of BMS. For example, Chen et al. [17] proposed a grey neural network particle filtering framework to predict the remaining useful life of lithium batteries. Wang et al. [18] proposed a grey prediction model of dynamic delay effect fraction system for lithium battery health state evaluation. However, these approaches do not explore the reasons for the use of grey model (GM) for lithium battery state estimation. The theory of grey system was first put forward by Deng in 1982 [19]. GM in grey system theory is a kind of single exponential model in essence, which agrees with the single exponential model in the empirical model to some extent. The use of empirical models alone generally does not result in accurate state estimation. It is typically used in combination with filtering type algorithms. For example, Lou et al. [20] fused a double-exponential empirical model and an extended Kalman filter algorithm for remaining useful life prediction of supercapacitors. Based on a review of the literature, the single-exponential model has been used in combination with unscented particle filtering [21] and particle filtering [22]. Meanwhile, the double exponential model has been jointly used with particle filtering [23–25], unscented particle filtering [26], and unscented Kalman filtering [27]. Additionally, the polynomial model has also been combined with Kalman filtering [28] and dual particle filtering [29]. This demonstrates that empirical models are mostly used in conjunction with filtering models.

Therefore, filtering algorithm and grey prediction model are integrated to improve the estimation accuracy in the research of state of charge estimation. Meanwhile, the GM contains the accumulation operator, which can transform the SOC data into an exponential type feature more easily to explore the degradation trend. Therefore, this paper designs the corresponding GM for the actual application scenarios of SOC.

This paper is divided into five sections, Section 1 is the current status of related theoretical research. Section 2 is the exposition of the SOC problem of lithium batteries. Section 3 shows the grey Kalman filtering model proposed in this paper. Section 4 shows the validation of the proposed model from different dimensions. Section 5 is about SOC estimation for electric vehicles. Section 6 is the conclusion of this paper.

2. Problem statement

In the failure prediction and health management of lithium batteries, the empirical model is one of the most classical models, in which the single-exponential model of the empirical model has a closer relationship with the GM. The formula of the single-exponential model is shown as follows:

$$y = a_1 e^{a_2 n} + a_3 \quad (1)$$

where a_1, a_2, a_3 are the parameters of the single-exponen-

tial model. n represents the cycle number of the lithium battery.

Taking the first order derivative of (1) and bringing (1) into the resultant derivative yields

$$\frac{dy}{dn} = a_2 y - a_2 a_3. \quad (2)$$

Let a_2 as $-a$ and $-a_2 a_3$ as b . While (2) is the original data sequence. Taking light vehicle drive cycle United States Government's light vehicle drive cycle (US06) as an example, the graph can be seen in Fig. 1.

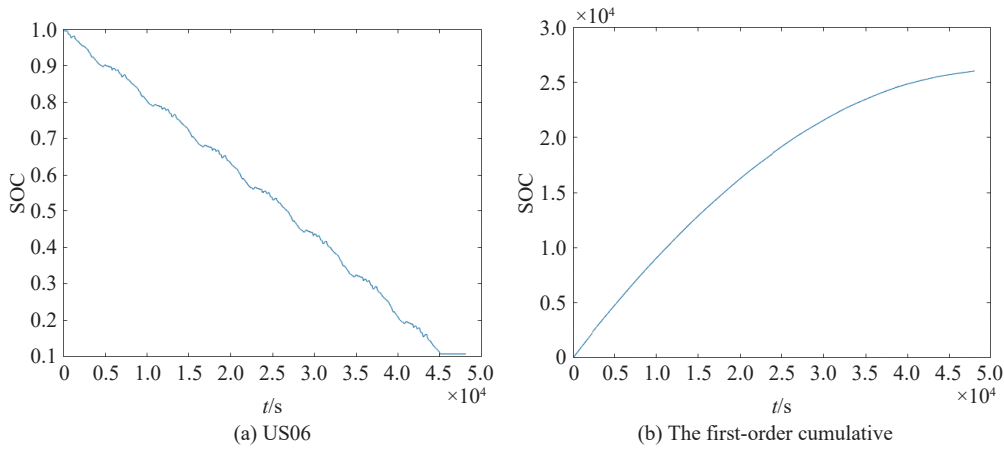


Fig. 1 US06 and its cumulative data

Fig. 1(a) is the raw data of the charge state of US06 under the driving cycle condition. There are many irregular turning points in Fig. 1(a), and it is challenging to characterize these points using empirical models. Fig. 1(b) is the sequence processed with the first-order cumulants. From the two graphs, it can be clearly seen that the processed sequence is much smoother and more trending. It can capture the SOC development trend more easily. This demonstrates the role played by the grey model. The GM is good at dealing with uncertain data and capturing the system trend by establishing the recursive relational equation of SOC. The Kalman filter dynamically corrects the prediction error, and the fusion of the two models achieves more accurate state estimation, especially under dynamic working conditions, and still maintains high accuracy.

The University of Wisconsin public lithium battery dataset is used in this paper. This section not only estimates SOC under different single driving cycles of lithium batteries but also estimates SOC values under multiple crossing of multiple driving cycles. In addition, considering that lithium batteries are affected by the tem-

perature environment, SOC values at different temperatures are estimated under the same driving cycle condition. Considering that this data is only from the laboratory and lacks application to real-world scenarios, SOC estimates for 20 EVs are added. The research idea of the grey Kalman filter model (GKFM) for estimating the state of charge of lithium batteries is shown in Fig. 2. In Fig. 2, k denotes the k th moment, $-a$ is the development coefficient in the GM, $\text{soc}^{(1)}(k)$ represents the first-order accumulated SOC value at the k moment. In the state-space model, $\text{soc}^{(0)}(k)$ denotes the restored SOC value at the k moment after inverse accumulation, w_k is the process noise, and v_k is the measurement noise.

The research steps are divided into the following five steps.

Step 1 Based on the inability of the empirical model to accurately estimate the SOC, a link between the empirical model and GM(1,1) is established, deriving the GM(1,1) model.

Step 2 The equations of state and observation are established according to GM(1,1).

Step 3 Combine the Kalman correlation theory to form a GKFM.

Step 4 SOC estimation is performed from laboratory data (University of Wisconsin public lithium battery dataset) by obtaining three dimensions: single driving cycle, mixed driving cycle, and different temperatures

under the same driving cycle.

Step 5 From laboratory data to real-life scenario case studies. The SOC estimation is performed based on real-life data from 20 EVs.

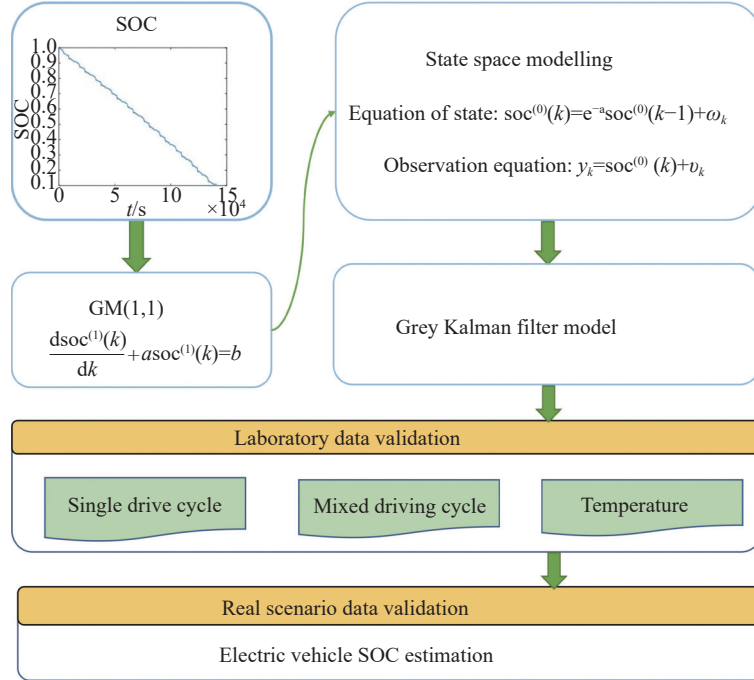


Fig. 2 Research idea of SOC estimation for lithium battery

3. GKFM

The model proposed in this paper is a combination of GM(1,1) and Kalman filter algorithm to form a GKFM to solve the lithium battery SOC estimation problem. The formation process of the model is shown as follows.

3.1 GM(1,1)

There is the raw data of lithium battery $\text{soc}^{(0)} = (\text{soc}^{(0)}(1), \text{soc}^{(0)}(2), \dots, \text{soc}^{(0)}(n))$. According to the definition of the GM(1,1) first-order cumulant operator, its first-order cumulant form is

$$\text{soc}^{(1)}(k) = \sum_{i=1}^k \text{soc}^{(0)}(i), \quad k = 1, 2, \dots, n \quad (3)$$

The basic form of the whitening differential equation for the GM(1,1) is given as follows:

$$\frac{d\text{soc}^{(1)}(t)}{dt} + a\text{soc}^{(1)}(t) = b. \quad (4)$$

Discretising (4) yields

$$\text{soc}^{(0)}(k) + az^{(1)}(k) = b \quad (5)$$

where

$$z^{(1)}(k) = \frac{1}{2}(\text{soc}^{(1)}(k) + \text{soc}^{(1)}(k-1)). \quad (6)$$

The parameter $\hat{a}$ and the parameter $\hat{b}$ can be estimated by least squares. $-\hat{a}$ is termed the development coefficient, $\hat{b}$ is referred to as the grey action quantity. $\mathbf{Y}, \mathbf{H}, \hat{\mu}$ are the output terms, as follows:

$$\hat{\mu} = (\mathbf{\Theta}^T \mathbf{\Theta}^{-1}) \mathbf{\Theta}^T \mathbf{Y} \quad (7)$$

$$\text{where } \mathbf{Y} = \begin{bmatrix} \text{soc}^{(0)}(2) \\ \text{soc}^{(0)}(3) \\ \vdots \\ \text{soc}^{(0)}(n) \end{bmatrix}, \mathbf{\Theta} = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \vdots & \vdots \\ -z^{(1)}(n) & 1 \end{bmatrix}, \hat{\mu} = \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix}.$$

Equation (4) yields a time-response equation can be obtained as follows:

$$\text{soc}^{(1)}(k+1) = \left(\text{soc}^{(0)}(1) - \frac{b}{a} \right) e^{-ak} + \frac{b}{a}. \quad (8)$$

This is a first-order cumulative sequence to get the prediction result, and its cumulative reduction formula as follows:

$$\text{soc}^{(0)}(k) = \text{soc}^{(1)}(k) - \text{soc}^{(1)}(k-1). \quad (9)$$

At this point, a curve on the charge state can already be obtained, in order to get further more accurate results. On the basis of GM(1,1), we use the GKFM to correct the results obtained. With the grey prediction model we can obtain the relationship between the k th term and the $(k+1)$ th term as follows:

$$\text{soc}(k) = e^{-a} \text{soc}(k-1). \quad (10)$$

3.2 Kalman filter model

The Kalman filtering method uses the estimated value of the previous moment to estimate the value of the next moment. The basic principle of Kalman filtering algorithm is as follows.

Let the system have state equations as

$$\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1} + \mathbf{B}\mathbf{u}_{k-1} + \boldsymbol{\omega}_{k-1} \quad (11)$$

where $\mathbf{x}_k$ is the state variable; $\mathbf{u}_k$ is the system input variable; $\mathbf{A}$ denotes the state transition matrix; $\mathbf{B}$ represents the input matrix; $\mathbf{Q}_c$ represents the covariance matrix; The process noise $\boldsymbol{\omega}_{k-1}$ follows a normal distribution $\boldsymbol{\omega} \sim N(0, \mathbf{Q}_c)$.

The observation equation is

$$\mathbf{y}_k = \mathbf{H}_m \mathbf{x}_k + \mathbf{v}_k \quad (12)$$

where $\mathbf{H}_m$ represents the observation matrix; $\mathbf{y}_k$ is the observed variable; the measurement noise $\mathbf{v}_k$ follows a normal distribution $\mathbf{v} \sim N(0, \mathbf{R}_c)$, $\mathbf{R}_c$ represents the covariance matrix.

The Kalman filtering algorithm mainly realizes the filtering process through two phases: prediction and updating.

Prediction phase: taking the state variable $\mathbf{x}_{k-1}$ and the system input variable $\mathbf{u}_{k-1}$ into the system state equation gives a preliminary estimate of the state variable at the k th moment.

$$\mathbf{x}_k = \mathbf{A}\mathbf{x}_{k-1} + \mathbf{B}\mathbf{u}_{k-1}. \quad (13)$$

Meanwhile, the error variance matrix $\mathbf{P}_k^-$ at the k th moment is estimated from the error variance matrix $\mathbf{P}_{k-1}$ at the $(k-1)$ th moment, which yields

$$\mathbf{P}_k^- = \mathbf{A}\mathbf{P}_{k-1}\mathbf{A}^T + \mathbf{Q}_c. \quad (14)$$

The correction coefficients are obtained by calculating the Kalman gain. $\mathbf{K}_k$ represents the Kalman gain.

$$\mathbf{K}_k = \frac{\mathbf{P}_k^- \mathbf{H}_m^T}{\mathbf{H}_m \mathbf{P}_k^- \mathbf{H}_m^T + \mathbf{R}_c}. \quad (15)$$

Update phase: the difference between the real-time observations and the system observations is corrected by using Kalman gain coefficients. At this point the measurement of the state variables is updated as

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{y}_k - \mathbf{H}_m \hat{\mathbf{x}}_k^-) \quad (16)$$

where $\hat{\mathbf{x}}_k^-$ represents the a priori state estimate at time k , $\hat{\mathbf{x}}_k$ represents the a posteriori state estimate at time k . Measurement of the error covariance is updated at the same time

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- \quad (17)$$

where $\mathbf{I}$ is the unit matrix. At this moment the Kalman's filtering process is realized after continuous prediction and correction. In this paper, we will get the state space expression through the grey time response equation, and integrate the Kalman filtering algorithm to realize the grey Kalman to estimate the charge state.

3.3 GKFM

For the estimation of lithium battery SOC, the existing technology directly adopts the empirical model to estimate which cannot reach the ideal accuracy. In this paper, we analyze GM(1,1) and the empirical model of lithium battery through derivation, and the grey accumulation operator is able to effectively reflect the cumulative use trend of the state of charge. Therefore, a grey Kalman filter model is designed to solve the lithium battery SOC estimation according to the actual problem, and a lithium battery charge state estimation method based on the GKFM is proposed.

From the above investigation, we can obtain the state and observation equations as follows, where ω'_k and v'_k are the state error and observation noise, respectively.

$$\text{soc}^{(0)}(k) = e^{-a} \text{soc}^{(0)}(k-1) + \omega'_k, \quad (18)$$

$$y'_k = \text{soc}^{(0)}(k) + v'_k, \quad (19)$$

where y'_k is the observation estimate. Let $x'_k = \text{soc}^{(0)}(k)$, represent the state variables at moment k ; $x'_{k-1} = \text{soc}^{(0)}(k-1)$ represent the state variables at moment $k-1$; $A' = e^{-a}$, A' is an element of matrix $\mathbf{A}$; Kalman filtering is mainly a process of estimating the lithium battery charging state through two phases: prediction and updating. In the prediction stage is the estimation of k moment state variable values using $k-1$ moment state variables, which is called the time update of state variables, which can be obtained as follows:

$$x'_k = A' x'_{k-1} \quad (20)$$

where x'_{k-1} represents the SOC value at moment $k-1$ and x'_k represents the SOC value that is estimated at moment k .

At the same time, the error variance P'_{k-1} at the $(k-1)$ th moment is used to estimate the error variance P_k^- at the k th moment, as shown as follows:

$$P_k^- = A'_k P'_{k-1} A'_k + Q'_c \quad (21)$$

where Q'_c is the variance value of ω_k .

On this basis, we can compute the Kalman gain K'_k to obtain the correction coefficients:

$$K'_k = P_k^- (P_k^- + R'_c)^{-1} \quad (22)$$

where R'_c is the variance value of v_k .

The second stage of Kalman estimation is the update stage. By using the difference between the data y'_k observed in real time at the k th moment and the predicted value, combined with the calculated Kalman gain coefficients K'_k in the GKFM model to correct the a priori estimation, the state variable at this point is updated as follows:

$$x'_k = \hat{x}'_k + K'_k(y'_k - \hat{x}'_k). \quad (23)$$

The error covariance matrix is also updated for the measurements.

$$P'_k = (1 - K'_k)P'^{-}_k \quad (24)$$

The SOC estimation evaluation metrics are measured using mean absolute percentage error (MAPE) and root mean square error (RMSE), which are calculated using the formulas shown as follows:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\widehat{\text{soc}}^{(0)}(i) - \text{soc}^{(0)}(i)}{\text{soc}^{(0)}(i)} \right|, \quad (25)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{soc}^{(0)}(i) - \widehat{\text{soc}}^{(0)}(i))^2}, \quad (26)$$

where n represents the total sample size, $\text{soc}^{(0)}(i)$ represents the true value, and $\widehat{\text{soc}}^{(0)}(i)$ represents the estimated value.

At this point, the GKFM construction is completed.

4. Estimated SOC for lithium batteries

In this paper, we use the University of Wisconsin public dataset to do comparative experiments under different driving cycles at 25 degrees Celsius, randomly mixed driving cycles, and experimental comparisons of different temperatures on the state of charge under the same driving cycle [30]. It is better to charge a lithium battery with about fifteen to thirty percent of its charge left, or one-third, because the process of fully charging a battery and the process of completely depleting it belongs to one cycle, and this percentage is the best charging cycle for lithium batteries. Lithium batteries should not be charged when there is excessive power loss, nor should they be charged when there is still a lot of power left in the battery, either of these operations will make the lithium battery become less durable, or even slowly unable to charge, at which point the battery will have to be replaced.

Under this premise, this paper preprocesses the data when the data is used, this paper discards SOC values below 30% to better estimate the effective state of charge

values and uses 1min as the node for data extraction to estimate the lithium battery SOC. 70% of the data length is used for the training parameters, and the remaining 30% of the training length is estimated. This subsection deals with lithium battery state of charge estimation in four dimensions, firstly lithium battery state of charge estimation under different driving cycles which include US06, highway fuel economy test (HWFET), UDDS, and Los Angeles 92 (LA92). Secondly lithium battery state of charge estimation under stochastic mixed driving cycle condition which contains four cycles, cycle 1–cycle 4 are from US06, HWFET, UDDS, and LA92 in a random mix. Thirdly, considering the variations in lithium batteries due to temperature, so to ensure the stability of the model, the state of charge values are estimated at different temperatures from the same driving cycle. Finally, the model is compared with other modelling algorithms as an illustration of the superiority of the model.

4.1 SOC estimation of lithium batteries under a single driving cycle

This subsection estimates the SOC of lithium batteries under different driving cycles, including US06, HWFET, UDDS, and LA92. Firstly, US06 is introduced as an example. US06 is an automotive emission test condition developed by the U.S. Environmental Protection Agency (EPA), which is designed according to the actual situation of traffic congestion and highway driving in the city and is used to evaluate the emission and fuel economy of the automobile under real conditions. Fig. 3 shows the estimated SOC under US06 conditions. In the process of estimating the state of charge, the parameter estimation yields a grey development coefficient a value of 0.014697, and the MAPE of the estimation result is 4.79%. the MAPE is generally used in a range lower than 10% to consider the model usable. In this paper, the estimation results are within the error range.

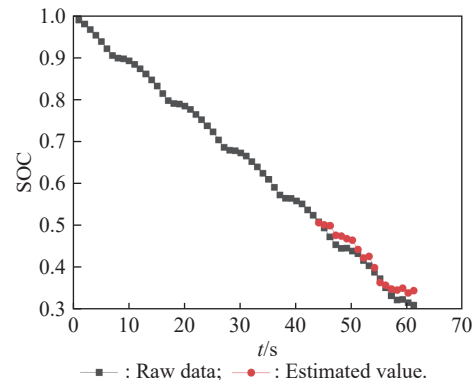


Fig. 3 SOC estimates at US06 driving cycles

Fig. 4 contains the estimated values under HWFET, UDDS, and LA92. HWFET is used to simulate performance at high speeds, and focuses on the fuel economy and emissions performance of the vehicle at high and steady speeds. The HWFET contains two sets of data, namely HWFETa and HWFETb. UDDS is different from US06 and HWFET, in that it is used to simulate the stop-and-go driving patterns of the city, especially frequent acceleration and deceleration, and is mainly used to test the fuel economy and performance of vehi-

cles under urban road conditions. LA92 focuses on simulating the driving conditions and characteristics of a specific city or region. The grey development coefficients of HWFETa, HWFETb, UDDS, and LA92 are 0.009486, 0.009498, 0.003104, and 0.004811, respectively. Their estimation errors are 3.05%, 2.71%, 4.61%, and 3.89%, respectively, and it can be seen in the display of the estimated error results that the proposed model is able to obtain good estimation results for a single driving cycle.

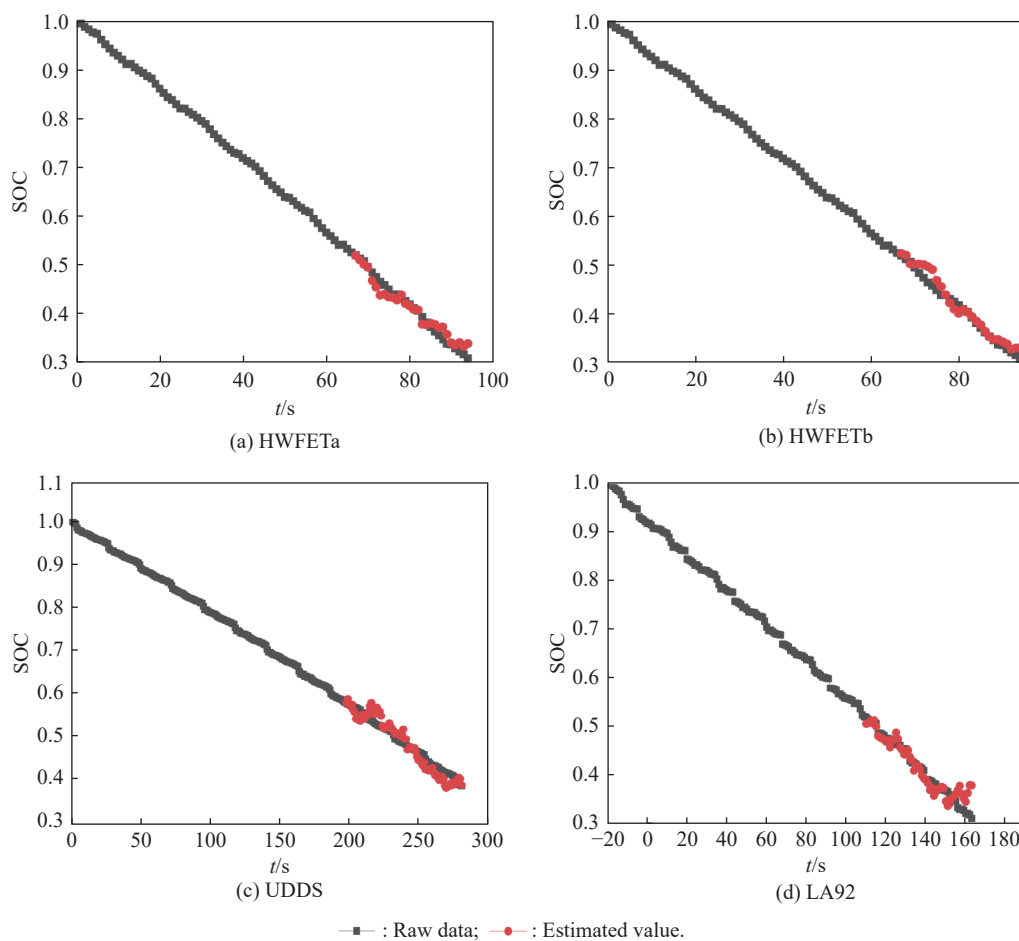


Fig. 4 SOC estimates for different driving cycles

4.2 SOC estimation of lithium batteries under stochastic mixed driving cycles

Subsection 4.1 is for the estimation of the lithium battery's state of charge under a single driving cycle condition. In the driving process of a real vehicle, the whole journey cannot be under the same working conditions all the time, to be closer to the use of real vehicles, this sub-

section adopts the stochastic mixed driving cycle data from the public dataset of the University of Wisconsin to estimate the state of charge of the lithium battery. The stochastic hybrid data contains four sets of data, including cycle 1, cycle 2, cycle 3, and cycle 4, which are random combinations of the different scenarios of US06, HWFET, UDDS, and LA92 mentioned in Subsection 4.1.

The results of the lithium battery charge state estimation under stochastic mixed driving cycle using the grey Kalman filtering model are shown in Fig. 5. In Fig. 5, it can be seen that the raw SOC data under the stochastic combined driving cycle starts to fluctuate more compared to that under a single operating condition, which shows that the SOC data under the stochastic combined driving cycle is indeed closer to the vehicle driving con-

dition. In this fluctuating state, the advantages of the proposed grey Kalman filtering model can be highlighted. The errors of the SOC estimates under random driving cycles are 5.03%, 3.51%, 4.75%, and 5.84%, respectively. These MAPE values are within a reasonable error range, so the performance of the GKFM can also be reflected from the perspective of SOC estimation under random mixed driving cycles.

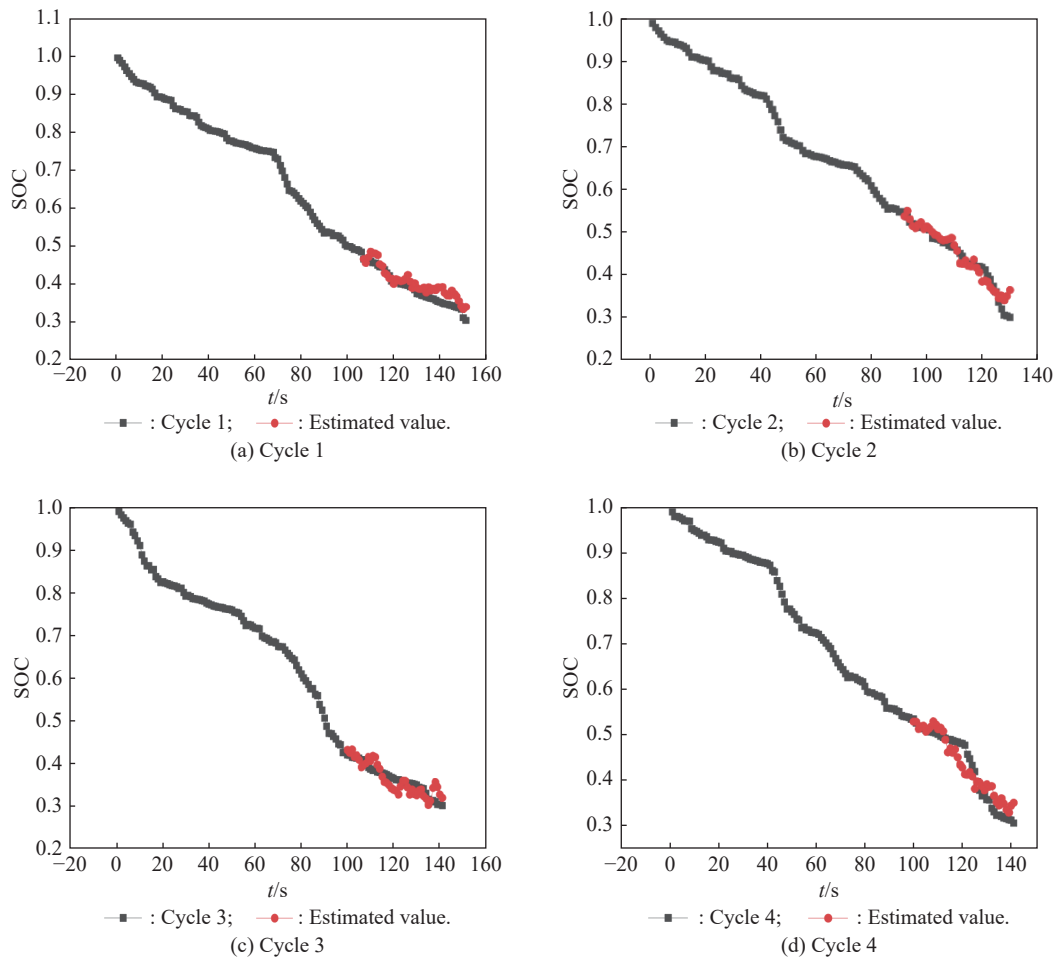


Fig. 5 SOC estimates under stochastic mixed driving cycles

4.3 SOC estimation of lithium batteries with different temperatures under the same driving cycle

Temperature variations affect the performance of lithium batteries to some extent. In this subsection, different temperature variations under the same driving cycle are used to simulate the estimation accuracy of SOC for the objectivity and fairness of the comparison test. Taking US06 as an example, the comparative analyses are conducted with different temperatures respectively, and the comparative

results are shown in Fig. 6. There are four temperature variations of US06 in the publicly available data, which are 10°C, 0°C, -10°C, and -20°C. The GKFM is used to estimate the SOC at the four temperatures with MAPE errors of 3.25%, 2.39%, 3.97%, and 2.77%, respectively. In this comparison result, it can be seen that with the change of temperature, the GKFM is still able to effectively estimate the SOC value of lithium battery, which further illustrates the effectiveness of the proposed model.

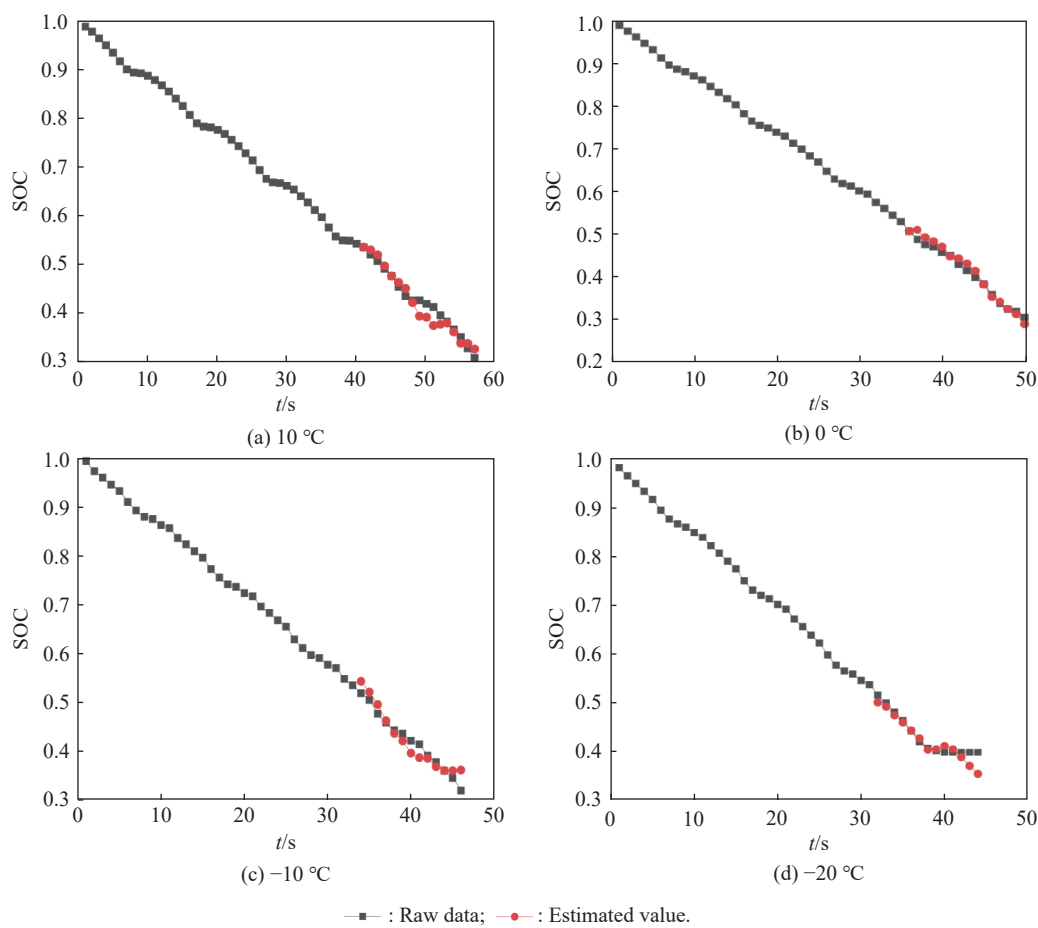


Fig. 6 SOC estimates with different temperatures

The robustness of the model can be reflected in the above SOC estimation accuracies under US06, HWFETa, HWFETb, UDDS, LA92 and mixed driving cycles. The wide adaptability of the model to the dynamic driving range is demonstrated. The multi-case validity predicts the potential application of the model in BMS, which is effective in estimating SOC for highway scenarios, urban commuting, and complex working conditions. Meanwhile, this also highlights the advantages of the GKFM. To further validate the applicability of the model in real-life scenarios, SOC estimation for 20 EVs is added in Section 5.

4.4 Comparison with other modelling algorithms

This Subsection will compare the GKFM with other models. Comparison models contain common grey single-variable prediction models such as GM(1,1), adjacent cumulative discrete grey GM(1,1) (ADGM). In addition to this, artificial intelligence algorithms such as support vector machines (SVM), neural network autoregressive (NNAR) model are also included. Table 1 shows the results of US06 under different comparison models. Among them, the order of ADGM is 0.1. NNAR and SVM are implemented by directly calling nnetar in the forecast package and SVM in e1071 in R language.

Table 1 SOC estimation results of US06 in different comparison models

k	US06	GKFM	GM	ADGM	NNAR	SVM
44	0.5062	0.504 0	0.5369	0.489 0	0.516 0	0.5523
45	0.4914	0.4989	0.529 0	0.4759	0.5108	0.5577
46	0.4703	0.497 0	0.5213	0.4626	0.5063	0.5653
47	0.4516	0.474 0	0.5137	0.4491	0.5024	0.5747
48	0.4427	0.4722	0.5062	0.4354	0.499 0	0.5858
49	0.4432	0.4659	0.4988	0.4215	0.4961	0.5981
50	0.4364	0.4623	0.4916	0.4074	0.4936	0.6113

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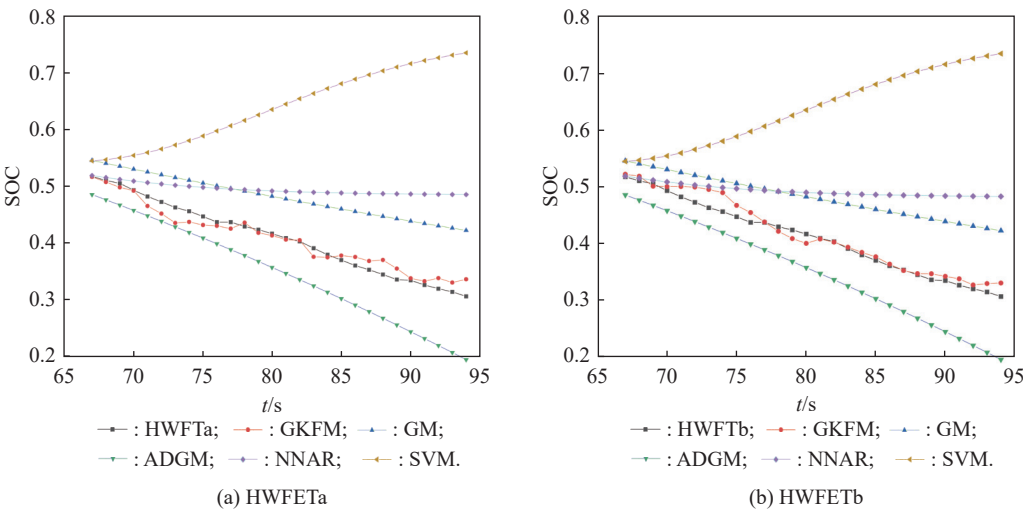
k	US06	GKFM	GM	ADGM	NNAR	SVM
51	0.430 0	0.440 1	0.484 4	0.393 1	0.491 5	0.625 1
52	0.413 7	0.419 7	0.477 3	0.378 5	0.489 6	0.638 9
53	0.401 2	0.423 6	0.470 4	0.363 8	0.488 1	0.652 6
54	0.385 6	0.396 7	0.463 5	0.348 7	0.486 8	0.665 9
55	0.370 3	0.361 2	0.456 7	0.333 5	0.485 6	0.678 4
56	0.348 6	0.354 7	0.450 1	0.318	0.484 7	0.689 9
57	0.329 4	0.345 6	0.443 5	0.302 3	0.483 9	0.700 5
58	0.319 4	0.343 2	0.437	0.286 4	0.483 2	0.709 9
59	0.320 8	0.347 7	0.430 7	0.270 2	0.482 7	0.718 2
60	0.313 0	0.336 2	0.424 4	0.253 8	0.482 2	0.725 3
61	0.307 1	0.341 8	0.418 2	0.237 1	0.481 8	0.731 4
RMSE	—	0.020 4	0.081 1	0.035 3	0.107 7	0.273 4
MAPE	—	0.047 9	0.207 4	0.084 8	0.262 7	0.679 1

The accuracy measures of the models are measured using MAPE and RMSE, respectively. As can be seen from Table 2, GKFM has the smallest error, which indicates that the proposed model outperforms the comparative models in terms of RMSE and MAPE.

Table 2 SOC estimator error under US06 driving cycle

Model	MAPE	RMSE
GKFM	0.047 9	0.020 4
GM	0.207 4	0.081 1
ADGM	0.084 8	0.035 3
NNAR	0.262 7	0.107 7
SVM	0.679 1	0.273 4

After comparing the SOC values of the US06 driving cycle, the authors also compared the models of HWFET, UDDS, and LA92 at the same temperature, and the comparison models are the same as US06, and the comparison results are shown in Fig. 7.



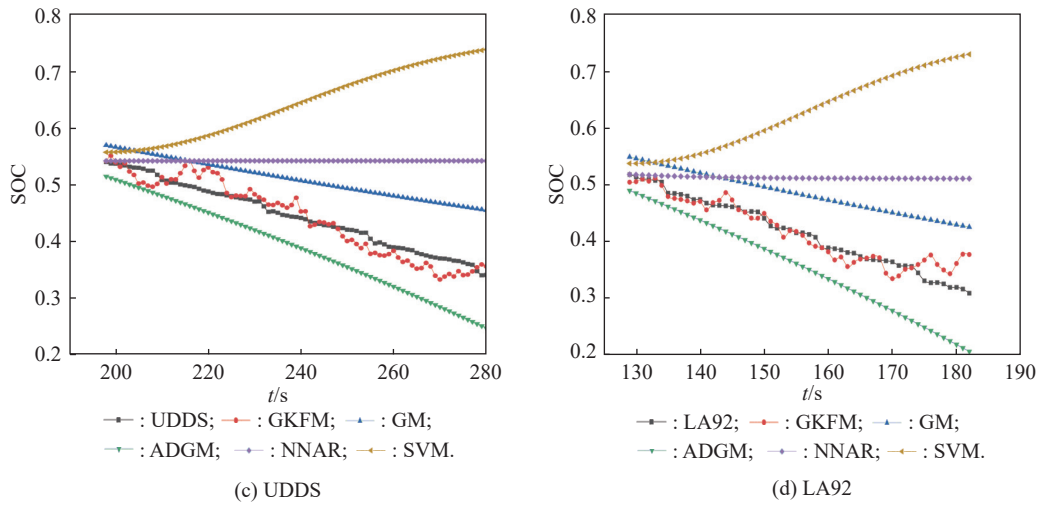


Fig. 7 The comparison results of SOC estimation

In the comparison model, GM(1,1) is the first-order cumulative GM(1,1). ADGM(1,1) is the adjacent cumulative GM(1,1), which has cumulative operator weights of 0.1 in HWFETa, HWFETb, UDDS, and LA92, and the search approach is to fit the model with a step size of GKFM0.1 from 0 to 1 to minimize the fitting error. NNAR GKFM and SVM are calculated from packages called directly GKFM in the R language. It can be seen from Fig. 7 that the GKFM can estimate the SOC value well.

In Table 3, it can be seen that the GKFM has the highest SOC estimation accuracy. At the same time, the SVM is the largest both in terms of RMSE and MAPE which error. Inside the grey comparison model, the improved ADGM(1,1) model is better than the classical GM(1,1) model, while the GKFM is better than the ADGM(1,1) model when doing SOC estimation for lithium batteries. This also shows the relevance and effectiveness of the improved grey prediction model for lithium battery SOC estimation.

Table 3 SOC estimates for different driving cycles

Model	HWFETa		HWFETb		UDDS		LA92	
	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
GKFM	0.030 5	0.014 4	0.027 1	0.014 5	0.046 1	0.000 5	0.038 9	0.020 3
GM	0.194 1	0.077 9	0.194 9	0.078 1	0.188 0	0.076 8	0.183 1	0.074 8
ADGM	0.162 2	0.065 4	0.162 4	0.065 4	0.160 4	0.065 6	0.124 9	0.063 8
NNAR	0.239 5	0.102 6	0.235 3	0.100 8	0.289 7	0.124 7	0.264 7	0.114 9
SVM	0.629 3	0.263 5	0.631 1	0.263 8	0.587 2	0.253 2	0.569 6	0.246 5

5. SOC estimation for electric vehicles

The dataset in this section is derived from the electric vehicle public dataset [31]. This dataset is different from the other datasets that come from lithium battery data measured in the laboratory, but is actual data from EVs. The dataset contains battery pack charging data from a total of 20 EVs. Each # x in the dataset represents data from one electric vehicle. The data spans over two years and approximately 29 months.

Fig. 8 is the raw data for the SOC of the #1 electric vehicle, which contains 854 591 pieces of data in Fig. 8. The horizontal axis in this graph is time and the vertical axis is SOC, with 100 indicating full charge and 0 indicating no charge. Since the amount of all SOC data is too large to easily see the characteristics of the data of #1

electric vehicle, the authors have extracted the data on the basis of #1 electric vehicle. This dataset is the charging data of the electric vehicle, and the SOC is according to a cyclic process such as insufficient power to charging, and power depletion in charging. Fig. 8 contains all the data of the SOC cycle process of #1 electric vehicle.

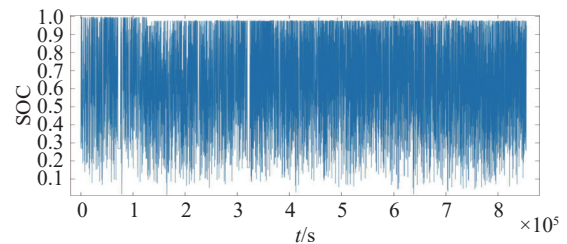


Fig. 8 SOC data for electric vehicle #1

Calculations are made according to the charging process, and the #1 electric vehicle is charged for a total of 1 566 cycles. In order to make it easier for readers to understand the individual charging cycle characteristics,

all the data of #1 EV are extracted by charging cycle, and now 1 cycle, the 500th cycle, the 1 000th cycle, and the last cycle of 1 566 are visualized, as shown in Fig. 9.

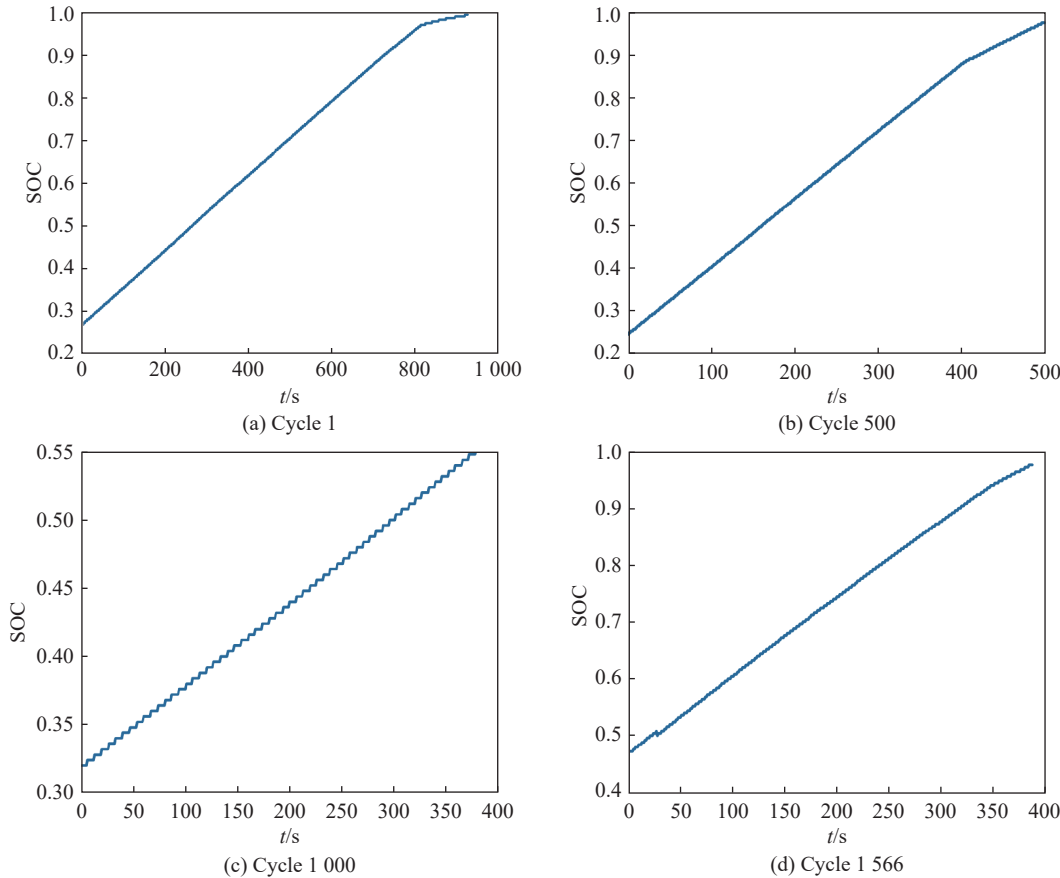


Fig. 9 Display of SOC data for electric vehicle #1

From Fig. 9 it can be seen that the electric vehicle battery pack is charged from a low charge to nearly full charge. At the same time, in the actual use of EVs, the lithium battery is not completely used until it has no power and then recharged or completely fully charged and then used, so it shows irregular up and down fluctuations in the complete charge state data. The charge cycle cycles are not exactly the same in the 20 EVs, and the statistics of the SOC charge cycle cycles of the 20 EVs are shown in Table 4. In Table 4 it can be seen that the charging cycle cycles of the 20 EVs vary greatly, with only 1 261 charging cycle data for electric vehicle number 15 and 1 846 cycles for electric vehicle number 13, indicating the differences in the frequency of use of different EVs. In Fig. 8, it can be seen that the lithium battery dataset for EVs is very large and further processing of the lithium battery dataset is required. In the existing dataset, take #1 electric vehicle as an example, there are 1 566 cycles, and the data length of each cycle varies. We

divided the data length into groups below 200, above 1 000, and between 200 and 1 000 with 50 as the interval point. The cycle sequence with the largest number of cycles is selected to find the mean value, which represents the SOC value of this electric vehicle. This results in a complete charging SOC sequence for 20 EVs. Under the premise of ensuring that the data trend is not changed, the sequence is spaced at intervals of 10 points, which can shorten the amount of data and retain the original trend of the SOC values.

Table 4 Charging cycle for electric vehicle SOC

EV	Cycle	EV	Cycle	EV	Cycle	EV	Cycle
#1	1 566	#6	1 504	#11	1 498	#16	1 559
#2	1 580	#7	1 518	#12	1 515	#17	1 404
#3	1 498	#8	1 509	#13	1 846	#18	1 549
#4	1 532	#9	1 455	#14	1 474	#19	1 469
#5	1 532	#10	1 322	#15	1 261	#20	1 499

In order to comprehensively evaluate the performance of the model on 20 EVs, we trained the data of these 20 vehicles separately and recorded the corresponding training parameters, which are shown in Table 5.

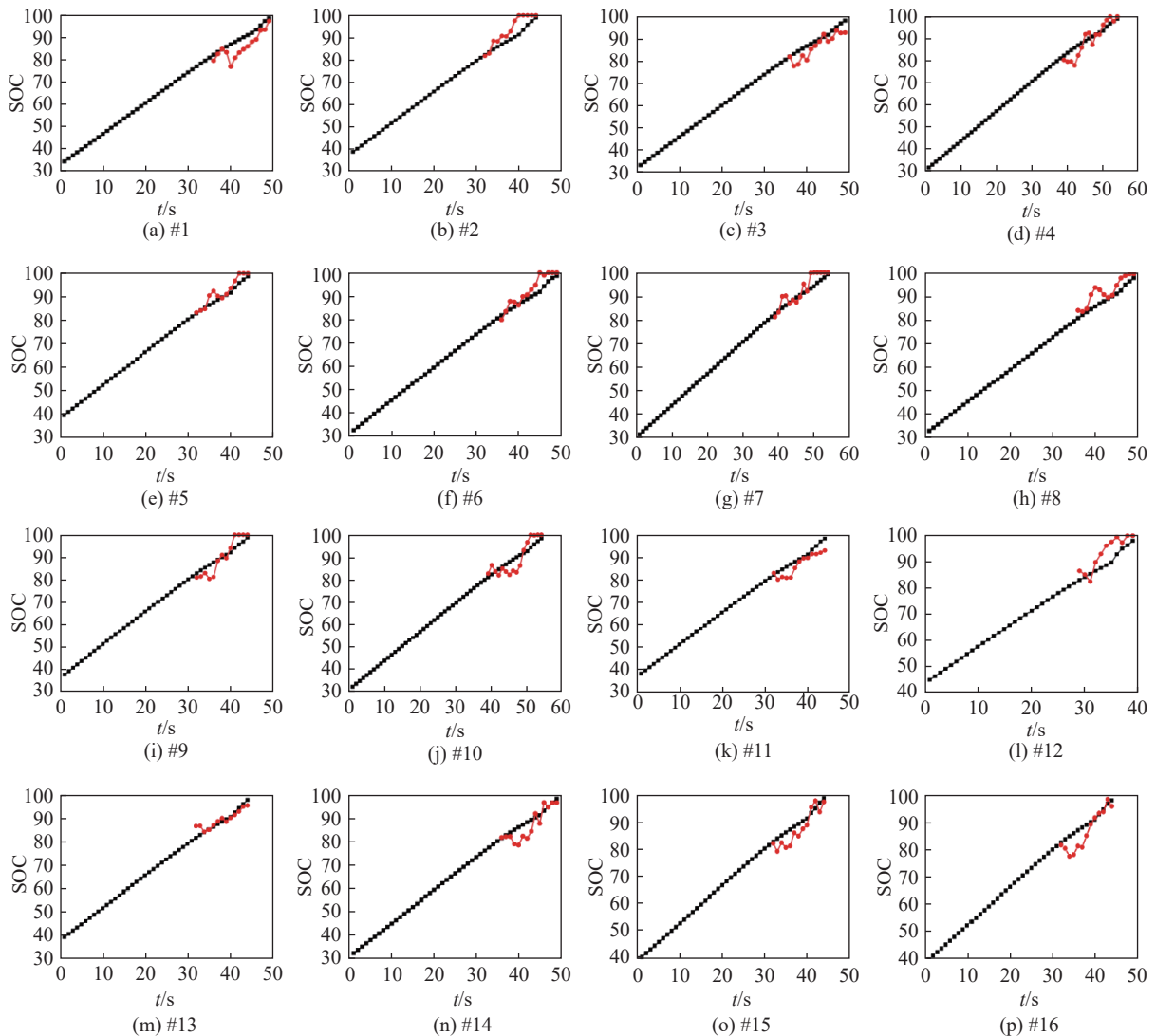
Table 5 Parameters for SOC estimation for EVs

EV	a	EV	a
#1	-0.0234	#11	-0.024
#2	-0.0231	#12	-0.021
#3	-0.0240	#13	-0.023
#4	-0.0233	#14	-0.024
#5	-0.0229	#15	-0.022
#6	-0.0245	#16	-0.023
#7	-0.0237	#17	-0.023
#8	-0.0240	#18	-0.024
#9	-0.0239	#19	-0.024
#10	-0.0228	#20	-0.025

On the basis of the obtained parameters, the SOC is estimated using the GKFM for 20 EVs, and the estima-

tion results are shown in Fig. 10. The graph visualizes the effect of the application of the model on different vehicles. In the SOC estimation of the lithium battery, the lithium battery is fully charged to 100, taking into account the actual situation, when the algorithm is designed, the value exceeding 100 is assigned to 100. such treatment not only ensures the accuracy of the estimation results, but also avoids the subsequent analysis and decision-making errors due to the abnormality of the estimated value.

According to the estimation results of lithium battery SOC, the proposed GKFM is able to estimate the SOC data of EVs, in which Fig. 10(m) and Fig. 10(q) almost coincide with the original data, which fully proves the excellent performance of the model in the complex practical application environment. By analyzing these two sets of data, the errors are calculated to be 1.77% and 1.72%, respectively, and their charge state estimation errors are well within the controllable range.



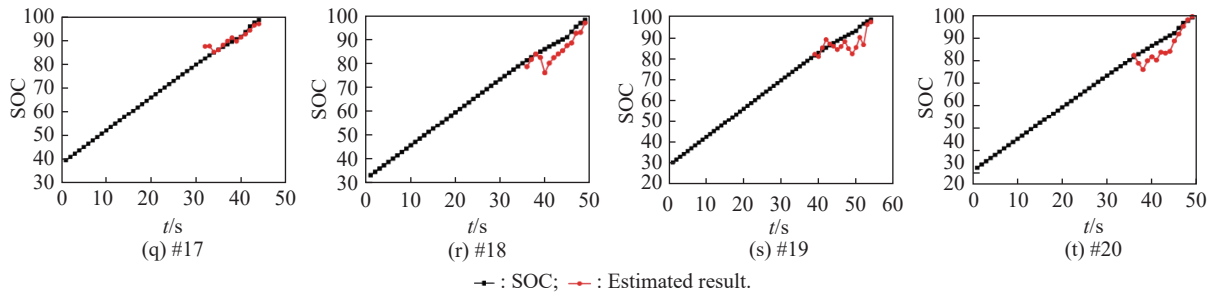


Fig. 10 SOC estimation results for 20 EVs

Table 6 summarizes the MAPE errors for the lithium battery SOC estimation. In Table 6 it is shown that the overall error of the lithium battery charge state estimation is basically stable at around 2% to 4%, which further indicates the stability of the model. The proposed model is able to perform good SOC estimation with the proposed grey Kalman filter model from the study of a single battery to the study of an electric vehicle battery pack, which further illustrates the applicability of the proposed model in practice.

Table 6 Errors in SOC estimation for EVs

EV	MAPE/%	EV	MAPE/%
#1	4.41	#11	3.43
#2	4.03	#12	4.64
#3	3.60	#13	1.77
#4	3.23	#14	3.53
#5	2.10	#15	3.60
#6	3.10	#16	3.79
#7	2.81	#17	1.72
#8	3.85	#18	4.35
#9	3.33	#19	4.23
#10	3.74	#20	4.67

The experimental results verify the universality of the GKFM in multi-scale applications. From the theoretical level, the GM implies the cumulative degradation trend of lithium battery SOC, which makes up for the insufficiency of the single-exponential empirical model. The proposed GKFM provides a priori estimation through the GM(1,1) model and corrects the measurement noise through Kalman filtering, which is good at dealing with non-stationary data under complex working conditions. In addition to this, the state equation established using the GM(1,1) model contains only one unknown parameter a , which further reduces the complexity of the model. From the engineering value level, the extension from individual battery to battery pack is realized. The accuracy of the model is verified in a individual battery through a single

driving cycle to a random mixed driving cycle, and the errors of the SOC estimates in cycle 1, cycle 2, cycle 3, and cycle 4 under a random driving cycle are 5.03%, 3.51%, 4.75%, and 5.84%, respectively. It can be seen that the model is still able to estimate the SOC of lithium battery well even under more complicated working conditions. Meanwhile, the practicality of the model is further verified in the estimation results of EVs, and the estimation error of the electric vehicle's state of charge numbered #17 can be up to 1.72%, which further illustrates the practicality of the model.

6. Conclusions

Since the vehicle may be more complex in the actual driving process, it is not possible to be in a single state all the time for electric vehicle driving, thus making it more difficult to accurately estimate the SOC of lithium batteries. In this article the authors propose a new GKFM for lithium battery SOC estimation by fusing the grey prediction model with the Kalman filter model. The model is also validated using publicly available lithium battery data from the University of Wisconsin-Madison. This dataset is a better case for use in SOC estimation, which contains a single working condition, and a random mixture of multiple working conditions. The stochastic mixture of working condition tests makes the simulation results closer to the real driving conditions of EVs. Based on this, we extended the model to real scenarios using 20 EVs SOC data. The following conclusions can be drawn in this paper.

(i) In this paper, the single-exponential model that exists through the empirical model in lithium batteries is expanded to obtain the GM(1,1) model in the grey system. Combining the grey system theory with the lithium battery theory, the GKFM is proposed, which further expands the GM architecture.

(ii) In this paper, the proposed GKFM is used for lithium battery SOC estimation, which further expands the application field of grey system theory. Compared with GM(1,1) and ADGM(1,1) models, this grey filter model specifically for SOC estimation of lithium batter-

ies is more advantageous than traditional grey prediction models.

(iii) In this paper, from the perspective of real driving of EVs, combined with the public dataset of lithium batteries, we estimate the SOC from multiple dimensions such as a single driving cycle, a randomly mixed driving cycle, and the effect of different temperatures on the SOC of the same driving cycle, to test the performance of the proposed model more accurately. Meanwhile, the validity of the model is further verified by using the data of EVs.

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